
STATS418 Proposal Presentation: Portfolio Risk Stress-Tester

Mark Wang
Department of Statistics & Data Science

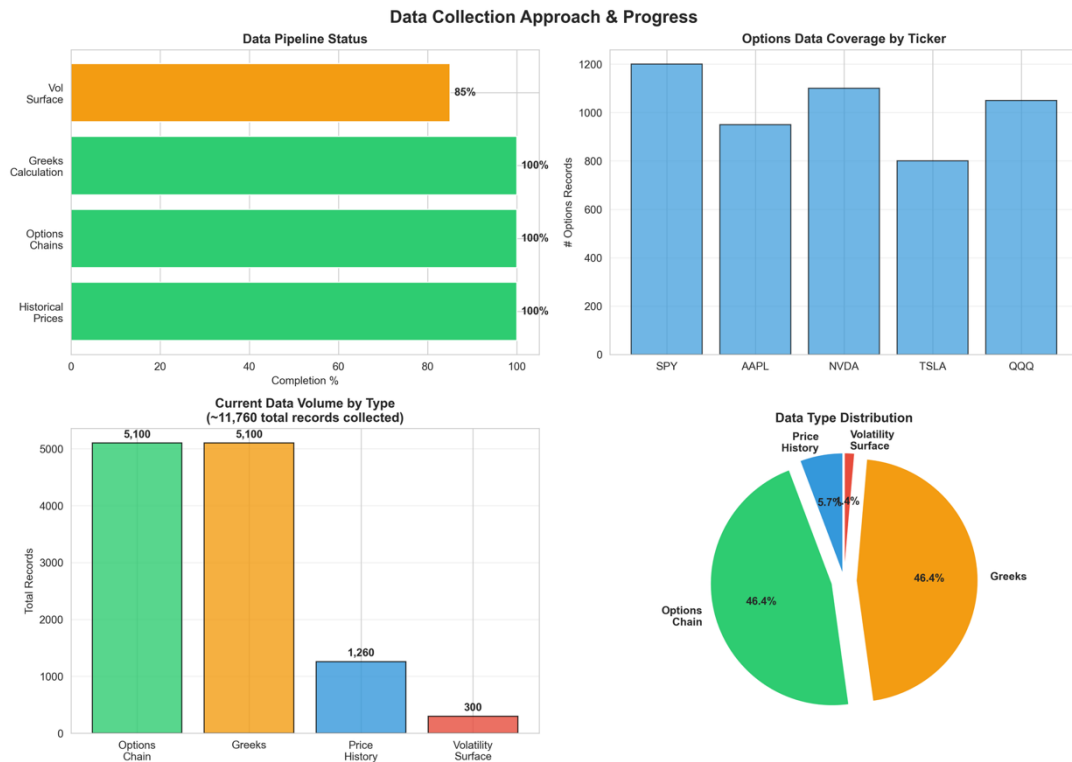
Motivation

PROPOSED APPLICATION

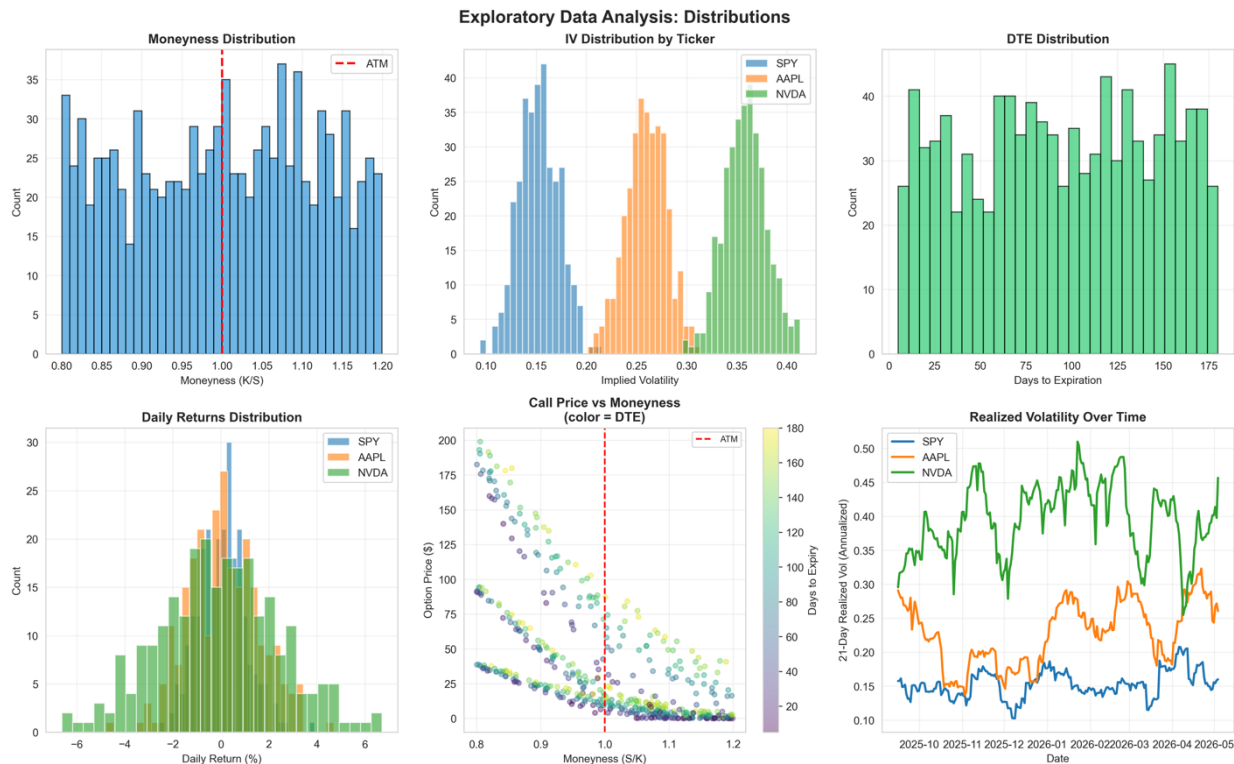
A large topic of focus in quantitative finance and risk management is the study of options trading. Thus, we are interested in building a system can do the following:

- Collect real market data (historical prices, options chains)
- Perform EDA to understand market structure
- Develop and compare multiple modeling approaches for options pricing
- Enable portfolio risk quantification through Value-at-Risk calculations

Data Collection Approach & Progress



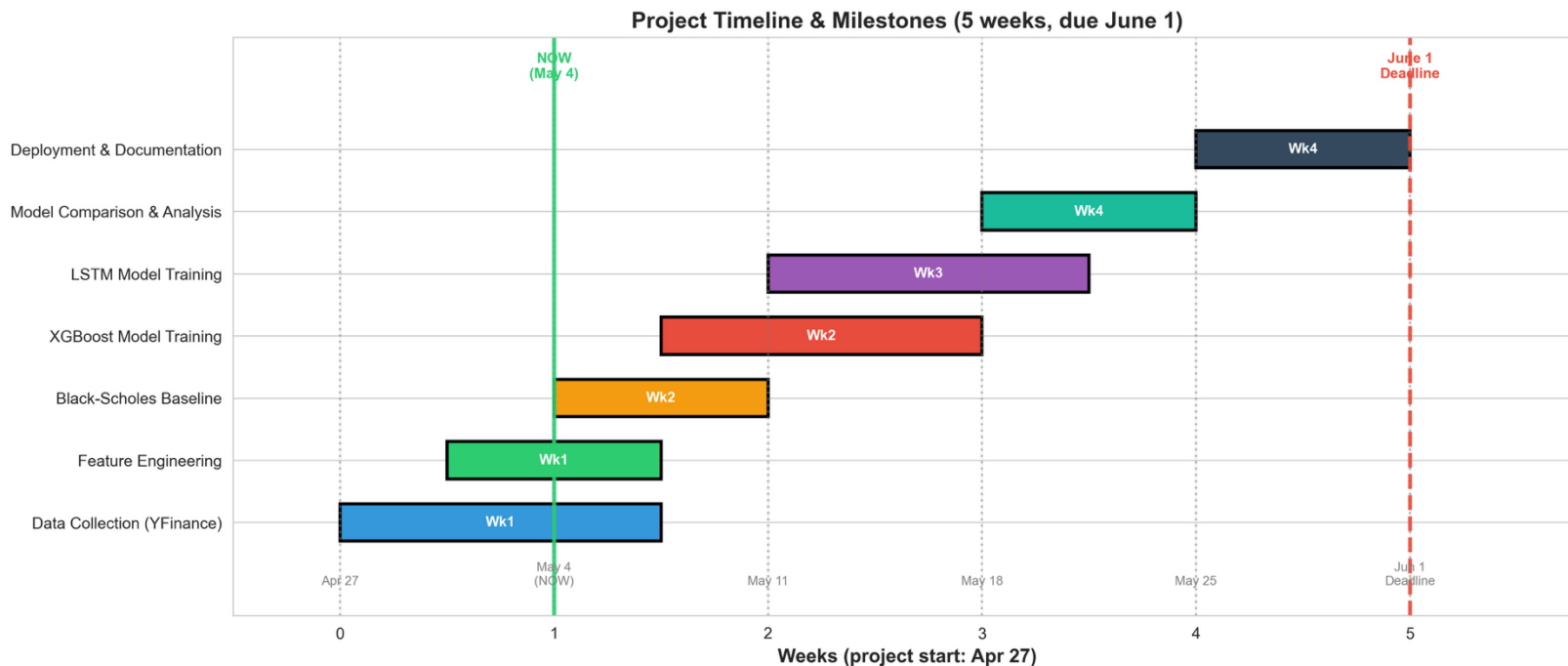
Exploratory Data Analysis



Proposed Modeling Approaches



Timeline & Milestones



Thank You
